

Propositional Logic Series

Episode 2: Worksheet Solutions

Part A. Truth-Apt Expressions

1. Truth-apt. It can be true or false.
2. Not truth-apt in the strict logical sense. It is a question.
3. Truth-apt. It makes a claim about integers.
4. Not truth-apt in the strict logical sense. It is a command or instruction.
5. Truth-apt. It can be evaluated as true or false.
6. Not truth-apt in the strict logical sense. It is an instruction.
7. Truth-apt. It makes a claim about sets.
8. Not truth-apt in the strict logical sense. It is an invitation or instruction.

Part B. Arguments and Nonarguments

1. Argument. The first two statements are presented as support for the conclusion that 120 is divisible by 3.
2. Not an argument. This is a list of truth-apt expressions. None of the statements is presented as support for another statement.
3. Argument. The statements $A \subseteq B$ and $x \in A$ are presented as support for the conclusion $x \in B$.
4. Not an argument. This is an instruction or problem command, not a passage in which premises are presented as support for a conclusion.
5. Not an argument in the given context. It functions as an explanation. The evenness of 48 is already accepted, and divisibility by 2 explains why 48 is even.
6. Argument. The first three statements are presented as examples supporting the general conclusion that every natural number divisible by 5 is divisible by 2.
7. Argument. The first two statements are presented as support for the conclusion that f is continuous at a .

8. Not an argument. This is a list of mathematical statements. No statement is presented as a conclusion supported by the others.
9. Argument. The first two statements are presented as support for the conclusion that the website is unavailable.
10. Not an argument in the given context. It functions as an explanation. The wetness of the ground is already observed, and the rain is offered as the reason why it is wet.

Part C. Premises and Conclusions

1. Premises: If an integer is divisible by 6, then it is divisible by 3; 120 is divisible by 6. Conclusion: 120 is divisible by 3.
2. Premises: Every differentiable function is continuous; f is differentiable at a . Conclusion: f is continuous at a .
3. Premises: 15 is divisible by 5; if a number is divisible by 5, then it has last digit 0 or 5. Conclusion: 15 has last digit 0 or 5.
4. Premises: $A \subseteq B$; $x \in A$. Conclusion: $x \in B$.
5. Premises: The server is down; if the server is down, the website is unavailable. Conclusion: The website is unavailable.
6. Premises: The triangle has three equal sides; every triangle with three equal sides is equilateral. Conclusion: The triangle is equilateral.
7. Premise: n is divisible by 4. Conclusion: n is even.
8. Premises: p is prime; $p > 2$. Conclusion: p is odd.

Part D. Standard Form

1. Premise: $72 = 8 \cdot 9$. Conclusion: 72 is divisible by 8.
2. Premises: $A \subseteq C$; $x \in A$. Conclusion: $x \in C$.
3. Premises: f is differentiable at a ; every differentiable function is continuous. Conclusion: f is continuous at a .
4. Premises: 36 is divisible by 6; every number divisible by 6 is divisible by 3. Conclusion: 36 is divisible by 3.
5. Premises: The file is corrupted; corrupted files cause the program to crash. Conclusion: The program crashes.
6. Premises: m is even; no even integer is odd. Conclusion: m is not odd.

Part E. Argument, Explanation, or Mere List

1. Mere list. Several truth-apt expressions are listed, but none is presented as supported by the others.
2. Explanation. The context says that we already know 48 is even and are asking why.
3. Argument. The premise that 48 is divisible by 2 is presented as support for the conclusion that 48 is even.
4. Explanation. The wet ground is already observed, and the rain is offered as the reason why it is wet.
5. Argument. The first two statements are presented as support for the conclusion $x \in B$.
6. Mere list. The statements are truth-apt, but no support relation is presented.

Part F. Deductive or Inductive

1. Deductive. The conclusion is presented as forced by the premises.
2. Inductive. The conclusion is a generalization from a few examples and is not forced by them.
3. Inductive. The conclusion generalizes from a few prime examples. The conclusion is false because 9 is odd and greater than 1 but not prime.
4. Deductive. If the premises are true, the conclusion is forced.
5. Inductive. The conclusion generalizes from the first ten values to all terms.
6. Deductive. The conclusion is presented as forced by the premises.
7. Inductive. The word “probably” signals that the support is not deductive.
8. Deductive. The conclusion is presented as forced by the premises.

Part G. Validity Preview

1. Valid. The first premise states that anything divisible by 4 is even. Since 20 is divisible by 4, it follows that 20 is even. There is no case in which the premises are true and the conclusion is false.
2. Invalid. The premises are true, but the conclusion is false. The integer 6 is even, but 6 is not divisible by 4. Thus, this is a case in which the premises are true and the conclusion is false.
3. Invalid. The first premise is true, and the function $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = |x|$ is continuous at 0. However, this function is not differentiable at 0. Thus, the premises are true and the conclusion is false.
4. Valid. If $A \subseteq B$, then every element of A is also an element of B . Since $x \in A$, it follows that $x \in B$. There is no case in which the premise is true and the conclusion is false.

Sources and Note

The conceptual background for these solutions is informed by P. D. Magnus, Tim Button, Robert Trueman, and Richard Zach, *forall x: Calgary: An Introduction to Formal Logic*, with contributions by J. Robert Loftis and Aaron Thomas-Bolduc.

Official page: <https://forallx.openlogicproject.org/>

The exercises and solutions are original teaching materials prepared for the Propositional Logic Series. They are not copied from the textbook.